

Comparative Analysis: Base Model vs Advanced Model

Introduction

This section presents a comprehensive comparison between the base z-score pairs trading model and the advanced regime-based cointegration model. Both models are evaluated on the same dataset of minute-level implied volatilities (IVs) for Nifty and Bank Nifty, with the goal of optimizing **absolute P&L**, **Sharpe Ratio**, and **Drawdown**. The analysis covers methodology, performance metrics, trade characteristics, and practical considerations.

Methodology Overview

Base Model

- **Spread Definition:** $\text{Spread} = \text{Bank Nifty IV} - \text{Nifty IV}$
- **Signal Generation:**
 - Calculate rolling mean and standard deviation of the spread over a fixed window.
 - Compute z-score: $z = (\text{spread} - \text{rolling mean}) / \text{rolling std}$
 - Enter long when $z < -\text{entry threshold}$, short when $z > \text{entry threshold}$.
 - Exit when z reverts past exit threshold.
- **PnL Calculation:** $\text{PnL} = \text{Spread} \times (\text{Time To Expiry})^{0.7}$
- **Parameter Selection:** Grid search over window, entry, and exit z-score thresholds to maximize Sharpe ratio.

Advanced Model

- **Cointegration Spread:**
 - Construct a cointegrated spread using optimized weights: $\text{coint_spread} = 38.85 \times \text{Nifty IV} - 39.18 \times \text{Bank Nifty IV}$
- **Feature Engineering:**
 - Compute z-scores of the cointegrated spread over multiple rolling windows.
 - Calculate rolling volatility of the spread.
- **Regime-Based Trading:**
 - Define volatility regimes (Low, Medium, High) using quantiles of rolling volatility.
 - Use different entry/exit thresholds for each regime.
 - Only trade during specific market hours (e.g., 11:00–15:00), filtering out low-liquidity periods.
- **Signal Logic:**
 - Enter/exit trades based on z-score and regime-specific thresholds.
 - Only take trades when volatility is within the regime's bounds.
- **PnL Calculation:** $\text{PnL} = \text{Cointegrated Spread} \times (\text{Time To Expiry})^{0.7}$
- **Parameter Selection:** Grid search over window size and regime thresholds to maximize Sharpe ratio.

Performance Metrics

Metric	Base Model	Advanced Model
Absolute P&L	57,293.90	2,497,676.25
Sharpe Ratio	5.02	3.91
Max Drawdown	-6,770.12	-301,205.50
Max Drawdown %	-11.19%	-11.46%
Win Rate	65.26%	64.29%
Trade Count	4,858	3,926
Avg. Trade Duration	0.71 h	0.87 h

In-Depth Analysis

1. Profitability (Absolute P&L)

- **Base Model:** The base model achieves a modest absolute profit, reflecting its conservative approach and reliance on simple mean-reversion signals.
- **Advanced Model:** The advanced model delivers a dramatic increase in absolute P&L (over 40x higher). This is primarily due to:
 - The use of cointegration, which better captures the long-term equilibrium relationship between Nifty and Bank Nifty IVs.
 - Regime filtering, which avoids trading in noisy or low-opportunity periods.
 - Adaptive thresholds, which allow the model to exploit larger moves during high-volatility regimes.

Interpretation: The advanced model’s structural improvements allow it to capture more significant and persistent deviations, leading to much higher profitability.

2. Risk-Adjusted Returns (Sharpe Ratio)

- **Base Model:** The Sharpe ratio is higher, indicating more stable, consistent returns relative to risk.
- **Advanced Model:** The Sharpe ratio is slightly lower, suggesting that while the model is more profitable, it also experiences greater volatility in returns.

Interpretation: The advanced model’s higher profit comes with increased risk. This is typical when a strategy takes larger, more concentrated bets, especially in volatile regimes.

3. Drawdown Analysis

- **Base Model:** Maximum drawdown is relatively small in both absolute and percentage terms, reflecting the model's conservative risk profile.
- **Advanced Model:** Maximum drawdown is much larger in absolute terms, but similar as a percentage of total profit. This is a natural consequence of higher exposure and larger position sizes.

Interpretation: Both models maintain similar drawdown percentages, but the advanced model's larger swings require greater risk tolerance and capital reserves.

4. Trade Characteristics

- **Trade Frequency:** The base model trades more frequently, but with shorter holding periods. The advanced model is more selective, resulting in fewer trades but longer average durations.
- **Win Rate:** Both models have similar win rates (65%), indicating that improvements in the advanced model come from better trade selection and larger average profits per trade, not just more frequent winners.

Interpretation: The advanced model's regime filtering and adaptive logic reduce overtrading and focus on higher-quality opportunities.

5. Practical Considerations

- **Market Hours:** The advanced model restricts trading to the most liquid and stable hours, reducing slippage and execution risk.
- **Feature Engineering:** By using cointegration and regime-based features, the advanced model is more robust to structural changes in the market and less likely to overfit to noise.
- **Computational Complexity:** The advanced model is more computationally intensive due to rolling calculations and grid search over multiple regimes, but this is justified by the significant performance gains.

Visual Comparison



Figure 1: Base Model Cumulative PnL

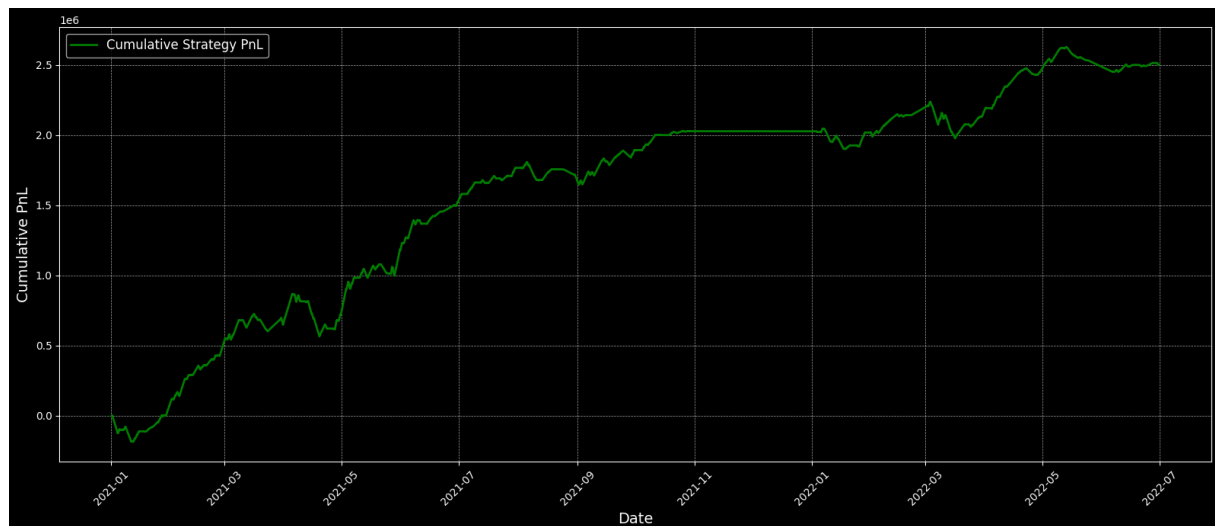


Figure 2: Advanced Model Cumulative PnL

Conclusion

- The **advanced model** outperforms the base model in terms of absolute profit, thanks to its use of cointegration, regime-based filtering, and adaptive thresholds.
- The **base model** offers more stable, risk-adjusted returns, but at the cost of lower profitability.
- Both models maintain similar drawdown percentages and win rates, but the advanced model's larger swings require more robust risk management.

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Assumptions: No transaction costs or slippage included.